

Week 2: Power Input Regulation - Theory Notes

Power Supply Design Essentials

These notes complement the lecture slides with detailed calculations, component selection guidance, and practical measurement techniques.

Efficiency and Power Dissipation

LDO Efficiency Formula

For a linear regulator, input current equals output current (plus small quiescent current):

$$\eta_{LDO} = \frac{V_{out}}{V_{in}} \times 100\%$$

Key insight: LDO efficiency is independent of load current - it depends only on voltage ratio.

Power Dissipation in LDO

$$P_{diss} = (V_{in} - V_{out}) \times I_{load} + V_{in} \times I_q$$

Where I_q is the quiescent current (typically 5-10 mA for AMS1117).

For most practical cases where $I_{load} \gg I_q$:

$$P_{diss} \approx (V_{in} - V_{out}) \times I_{load}$$

Buck Converter Efficiency

Buck efficiency varies with load current and is typically given as a curve in datasheets. Approximate formula:

$$\eta_{buck} = \frac{V_{out} \times I_{out}}{V_{in} \times I_{in}} \times 100\%$$

Typical values for LM2596:

Load Current	Efficiency (12V→3.3V)
50 mA	75-80%
100 mA	80-85%
200 mA	83-88%
500 mA	85-90%

Thermal Design Calculations

Junction Temperature

$$T_j = T_a + (P_{diss} \times \theta_{ja})$$

Parameter	Description	Typical Values
T_j	Junction temperature	Max 125°C
T_a	Ambient temperature	25-40°C
θ_{ja}	Thermal resistance (junction to ambient)	90°C/W (SOT-223)
θ_{jc}	Thermal resistance (junction to case)	15°C/W

Maximum Power Dissipation

Solving for maximum allowed power:

$$P_{max} = \frac{T_{j,max} - T_a}{\theta_{ja}}$$

Example: AMS1117 at 40°C ambient:

$$P_{max} = \frac{125^{\circ}C - 40^{\circ}C}{90^{\circ}C/W} = 0.94W$$

Adding a Heatsink

With heatsink, effective thermal resistance decreases:

$$\theta_{total} = \theta_{jc} + \theta_{cs} + \theta_{sa}$$

Where:

- θ_{jc} = junction to case (from datasheet)
 - θ_{cs} = case to sink (thermal pad/grease, ~0.5°C/W)
 - θ_{sa} = sink to ambient (heatsink rating)
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Component Selection

AMS1117-3.3 Key Specifications

Parameter	Min	Typical	Max	Unit
Input voltage	-	-	15	V
Output voltage	3.267	3.3	3.333	V
Dropout voltage (800mA)	-	1.0	1.3	V
Output current	-	-	1000	mA
Quiescent current	-	5	10	mA
Line regulation	-	0.2	0.4	%/V
Load regulation	-	0.4	0.8	%
Thermal resistance (SOT-223)	-	90	-	°C/W

Capacitor Selection for AMS1117

Input capacitor:

- Minimum: 10µF
- Recommended: 22µF ceramic (X5R or X7R)

- Purpose: Bypass high-frequency noise, prevent input oscillation

Output capacitor:

- Minimum: 22 μ F with ESR 0.1-0.5 Ω
- Recommended: 22-47 μ F ceramic
- Purpose: Stability, transient response

Important: Some LDOs require specific ESR range for stability. Always check datasheet!

Ceramic Capacitor DC Bias Derating

Ceramic capacitors lose capacitance under DC bias. A “22 μ F” capacitor at 3.3V may only provide 15 μ F effective capacitance.

Rule of thumb: Use capacitors rated for $2\times$ your operating voltage, or use a larger value.

Powering the MCU: Decoupling and Layout

Why Decoupling Capacitors Are Essential

The voltage regulator provides stable DC voltage, but the MCU doesn’t draw constant current. Every clock cycle, transistors switch and create current transients.

The problem:

- MCU current changes by 10-100 mA in nanoseconds
- Wire and trace inductance opposes rapid current change
- Voltage at MCU pins drops during current spikes

The physics:

$$V_{drop} = L \times \frac{di}{dt}$$

Even 10 nH of trace inductance with a 50 mA change in 10 ns causes:

$$V_{drop} = 10 \times 10^{-9} \times \frac{0.05}{10 \times 10^{-9}} = 0.5V$$

A 0.5V drop on a 3.3V supply is 15% - enough to cause logic errors!

Decoupling Capacitor Function

Decoupling capacitors act as local charge reservoirs:

1. **During current spike:** Capacitor supplies current, MCU voltage stays stable
2. **Between spikes:** Regulator recharges capacitor
3. **High-frequency noise:** Capacitor provides low-impedance path to ground

Key insight: The capacitor must be close to the MCU because every millimeter of trace adds inductance.

Capacitor Selection for Decoupling

100nF (0.1μF) ceramic capacitors:

- Standard bypass capacitor for digital ICs
- Low ESR and ESL in 0402/0603/0805 packages
- Effective frequency range: 1 MHz to 100 MHz
- Self-resonant frequency around 30-50 MHz (package dependent)

Bulk capacitors (4.7μF - 10μF ceramic):

- Handle lower-frequency transients
- Supplement regulator output capacitor
- Effective frequency range: 10 kHz to 10 MHz

Capacitor dielectric types:

Type	Capacitance Stability	Use Case
C0G/NP0	±30 ppm/°C	Precision, timing
X7R	±15% over temp	General bypass
X5R	±15% over temp	General bypass
Y5V	+22%/-82%	Avoid for bypass

Important: X7R and X5R lose capacitance under DC bias. A 10μF capacitor at 3.3V may only provide 6-7μF effective capacitance.

STM32F401 Power Supply Requirements

From the STM32F401 datasheet:

Supply	Voltage Range	Purpose
VDD	1.7V - 3.6V	Digital core, I/O
VDDA	1.7V - 3.6V	Analog (ADC, PLL)
VBAT	1.65V - 3.6V	RTC, backup registers

Pin count (LQFP64 package):

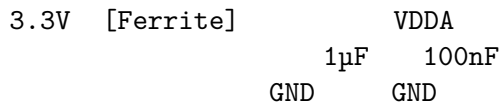
- 4× VDD pins (connect all!)
- 4× VSS pins
- 1× VDDA pin
- 1× VSSA pin
- 1× VBAT pin

Recommended Decoupling Scheme

Per STM32F401 hardware design guidelines:

Location	Capacitor	Notes
Each VDD-VSS pair	100nF ceramic	Place within 5mm of pins
One VDD-VSS pair	4.7μF ceramic	Bulk storage
VDDA-VSSA	1μF + 100nF	Extra filtering
VBAT	100nF	If using RTC

VDDA filtering for clean ADC:



The ferrite bead (10-100Ω @ 100MHz) blocks high-frequency noise from digital circuits.

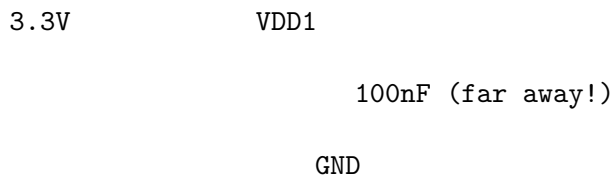
PCB Layout Guidelines

Critical placement rules:

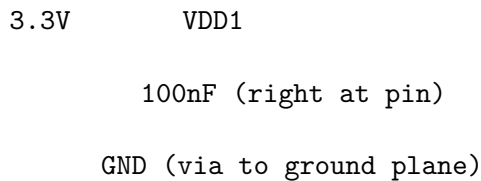
1. **Capacitor proximity:** Decoupling caps must be as close as possible to VDD/VSS pins (ideally < 3mm)

2. **Via placement:** If capacitor is on bottom layer, place via between capacitor and MCU pin, not beyond the capacitor
3. **Ground plane:** Use solid ground plane on inner layer; ground return path matters as much as power path
4. **Trace width:** Power traces should be wide (20-50 mil for main power, 10-20 mil for individual VDD)
5. **No daisy-chain:** Don't route power pin-to-pin through MCU; use star topology from main rail

Bad layout:



Good layout:



Common Power Problems and Solutions

Symptom	Likely Cause	Investigation	Solution
Random resets	Brown-out during current spikes	Check VDDA with scope during GPIO activity	Add bulk cap, improve layout
ADC readings noisy	VDDA noise	Scope VDDA during conversions	Add ferrite bead, separate VDDA ground
Unreliable at high speed	Power integrity issues	Check all VDD pins with scope	Verify all caps present and close
Works on bench, fails in enclosure	Thermal issues affecting caps	Check temperature rise	Use X7R instead of Y5V

Power-On Sequencing

STM32F401 requirements:

- VDDA must be VDD (can be equal)
- VDD and VDDA should rise together or VDDA first
- VBAT can be powered before VDD

For simple designs: Connect VDDA to VDD through ferrite bead; power rises together naturally.

For multi-rail designs: May need sequencing IC or careful regulator enable timing.

Load Resistor Calculations

For testing regulators, calculate load resistor values:

$$R_{load} = \frac{V_{out}}{I_{load}}$$

$$P_{resistor} = V_{out} \times I_{load} = \frac{V_{out}^2}{R_{load}}$$

Example: 3.3V output, 200mA load:

$$R_{load} = \frac{3.3V}{0.2A} = 16.5\Omega$$

$$P_{resistor} = 3.3V \times 0.2A = 0.66W$$

Use at least 1W resistor for safety margin.

Target Current	Resistor Value	Power Rating
50 mA	66 Ω	0.25W
100 mA	33 Ω	0.5W
200 mA	16.5 Ω	1W
300 mA	11 Ω	1W

Oscilloscope Measurement Techniques

Ripple Measurement Setup

Critical settings:

1. **Coupling:** AC (blocks DC, shows only AC ripple)
2. **Bandwidth limit:** 20 MHz ON (reduces high-frequency noise pickup)
3. **Probe ground:** Use spring ground, not long alligator clip
4. **Voltage scale:** Start at 50 mV/div, adjust as needed

Ground Loop Prevention

Long ground leads act as antennas, picking up noise. This creates false ripple readings.

Correct technique:

- Use scope probe's spring-loaded ground tip
- Or solder a short wire to ground plane
- Keep ground lead under 2 cm

Measuring LDO Ripple

LDO ripple is typically very low (< 10 mV). You may see mostly noise floor.

Expected waveform: Random low-level noise, possibly 120 Hz component from rectified AC.

Measuring Buck Ripple

Buck ripple shows clear switching frequency component.

LM2596 characteristics:

- Switching frequency: 150 kHz
- Expected ripple: 30-100 mV pk-pk
- Waveform: Sawtooth or triangular shape

Set timebase to show 5-10 switching cycles:

$$T_{period} = \frac{1}{150kHz} = 6.67\mu s$$

Use 5-10 μs /div timebase.

Efficiency Measurement Procedure

Step-by-step Method

1. **Set up power supply:**
 - Voltage: 12V
 - Current limit: 500 mA
 - Display both V and I
2. **Connect ammeter in series with output:**
 - Measure actual load current
 - Or use calculated R and measured Vout
3. **Calculate efficiency:**

$$\eta = \frac{V_{out} \times I_{out}}{V_{in} \times I_{in}} \times 100\%$$

Measurement Table Example

Parameter	Symbol	LDO Measured	Buck Measured
Input voltage	V_{in}	12.00 V	12.00 V
Input current	I_{in}	205 mA	78 mA
Output voltage	V_{out}	3.28 V	3.31 V
Output current	I_{out}	200 mA	200 mA
Input power	P_{in}	2.46 W	0.94 W
Output power	P_{out}	0.66 W	0.66 W
Efficiency	η	26.8%	70.2%

Troubleshooting Guide

LDO Issues

Symptom	Possible Cause	Solution
No output	Reversed polarity	Check pin orientation
Low output	Vin below dropout	Increase Vin > Vout + 1.3V
Oscillation	Wrong capacitor	Use 22μF ceramic, check ESR
Overheating	Current too high	Reduce load, add heatsink
Output noise	Missing input cap	Add 10μF ceramic on input

Buck Module Issues

Symptom	Possible Cause	Solution
No output	Vin below minimum	Need Vin > 4.5V for LM2596
Wrong voltage	Pot adjustment	Turn potentiometer, verify with meter
High ripple	Output cap failure	Check/replace output capacitor
Audible whine	Light load	Normal at very low currents
Efficiency low	High ESR capacitors	Use low-ESR ceramics

Quick Reference

Voltage Regulator Selection Guide

Condition	Recommended	Reason
Vin - Vout < 2V	LDO	Efficient enough
Vin - Vout > 3V	Buck	LDO too inefficient
Noise-sensitive load	LDO	Lower ripple
Battery powered	Buck	Longer runtime
Cost priority	LDO	Simpler, cheaper
High current (> 500mA)	Buck	Less heat

Common Regulator ICs

Part Number	Type	Vout	I _{max}	Notes
AMS1117-3.3	LDO	3.3V	1A	Most common
LM7805	Linear	5V	1A	Classic, high dropout
LM2596	Buck	Adj	3A	Popular module
MP1584	Buck	Adj	3A	More efficient
AP2112K	LDO	3.3V	600mA	Low quiescent

Formula Reference

Formula	Description
$\eta_{LDO} = V_{out}/V_{in}$	LDO efficiency
$P_{diss} = (V_{in} - V_{out}) \times I$	LDO power dissipation
$T_j = T_a + P \times \theta_{ja}$	Junction temperature
$D = V_{out}/V_{in}$	Buck duty cycle
$R = V/I$	Load resistor
$P = V \times I = V^2/R$	Power in resistor